

Supplementary Information

Landmark-specific transcriptomic and proteomic associations with 60-day mortality in Day-1-defined sepsis-induced coagulopathy: a multicentre longitudinal cohort study

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Supplementary Methods

Molecular data generation in the parent cohort

The deposited molecular matrices were generated under the CMAISE laboratory workflow. Whole-blood samples were collected in PAXgene tubes and processed after ribosomal-RNA depletion using a strand-specific dUTP/UDG library protocol; libraries were sequenced on an Illumina NovaSeq 6000 platform with 2×150-bp paired-end reads. Reads were adapter- and quality-filtered with cutadapt v1.9, assessed with FastQC v0.10.1, aligned to Ensembl GRCh38 v96 with HISAT2 v2.0.4, and assembled and quantified with StringTie v1.3.4, with GffCompare used to reconcile annotations. Plasma underwent magnetic-nanobead enrichment of low-abundance proteins, tryptic digestion, C18 desalting and 13-minute liquid-chromatography separation before positive-ion data-independent acquisition on an Orbitrap Astral mass spectrometer. DIA-NN v1.9.2 was used in DirectDIA mode with target-decoy control and peptide-spectrum-match and protein false-discovery rates below 1%. These procedures generated the deposited matrices; the present study then applied the longitudinal downstream processing described below.

RNA-seq processing

For the present longitudinal reanalysis, we started from the deposited integer RNA-seq count matrix rather than the variance-stabilised or TPM-derived matrices used for the parent baseline analyses. Counts were restricted to the 1,246 longitudinal SIC RNA-seq samples. Features were retained when they reached at least one count per million in at least 20% of longitudinal samples. Ensembl identifiers were harmonised to gene symbols using the frozen annotation map, and records mapping to the same symbol were combined by summing counts. Library sizes and log counts per million were estimated jointly across all three sampling landmarks using trimmed-mean-of-M-values normalisation rather than normalising each landmark independently. For each time-specific Cox analysis, expression of every retained gene was standardised within the corresponding risk-valid dataset; hazard ratios therefore represented a one-standard-deviation increase in expression.

Plasma protein processing and quality control

For the present reanalysis, the supplied quantitative plasma protein matrix was evaluated on its deposited log₂ scale and showed no material missingness; no K-nearest-neighbour or left-censoring imputation was applied. Seven rows without a gene symbol were removed. Before feature filtering, the three duplicated symbols (CUX1, RABGAP1L and TMPO) were resolved by retaining the UniProt entry with the highest median abundance across the 431 matched clinical proteomic samples, with original row order used only as a deterministic tie-breaker. Replicate variability was assessed across ten QC batches with three replicates per batch. Within-batch standard deviations on the deposited log₂ scale were converted to the corresponding linear-scale coefficients of variation, and the median coefficient of variation across batches was used for filtering. All 5,614 unique symbols passed the clinical-sample missingness criterion; 3,875 proteins with a median replicate coefficient of variation below 20% entered the final time-specific prognostic analyses. Protein abundance was standardised within each risk-valid dataset. A prespecified sensitivity analysis additionally adjusted for sample median protein intensity to evaluate global-intensity effects.

Day-5 availability weighting implementation

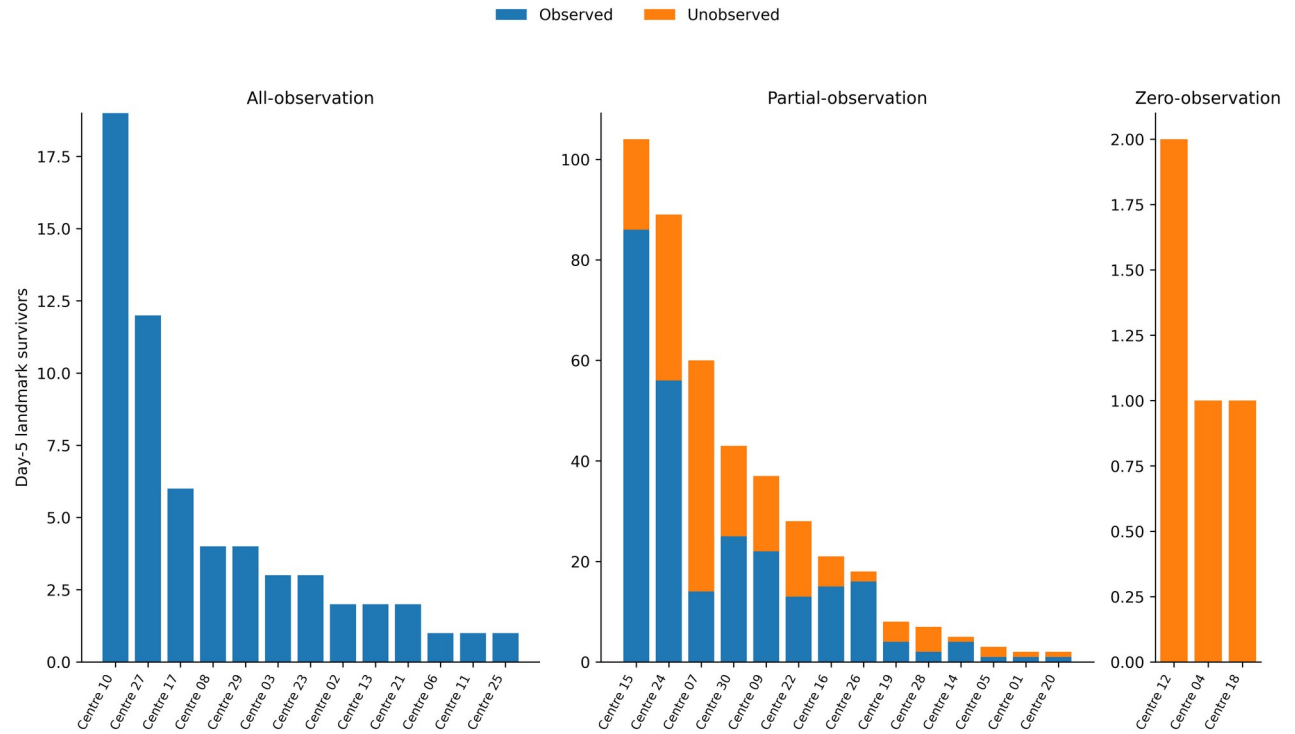
The prespecified primary observation model included baseline age, sex, SOFA score, infection source, centre, prior Day-3 RNA-seq availability, and lactate, INR and D-dimer, each log₂-transformed before

standardisation. Non-stabilised inverse observation probabilities were calculated as the reciprocal of the estimated probability, truncated at the 1st and 99th percentiles and normalised to mean one among observed patients. Weighted Day-5 delayed-entry Cox coefficients were estimated from the same centre-stratified partial likelihood. Observation-level Lin–Wei sandwich variances were calculated from weighted dfbeta residuals and independently verified against a model assigning a unique cluster to every row. All 14,541 successfully estimated genes were ranked by the weighted robust Wald statistic for Hallmark enrichment. Weighted and unweighted pathway profiles were compared across the complete Hallmark family using NES Spearman correlation, pathway-direction agreement and Jaccard overlap of FDR-significant sets. The primary weighted analysis was prespecified as a sensitivity analysis and was not intended to replace the unweighted centre-stratified result. Alternative observation models and entry-boundary scenarios were used only as prespecified sensitivity analyses. The weighted analysis evaluated robustness only within the positivity-supported landmark-survivor population and was not interpreted as complete correction for centre-structured non-observation.

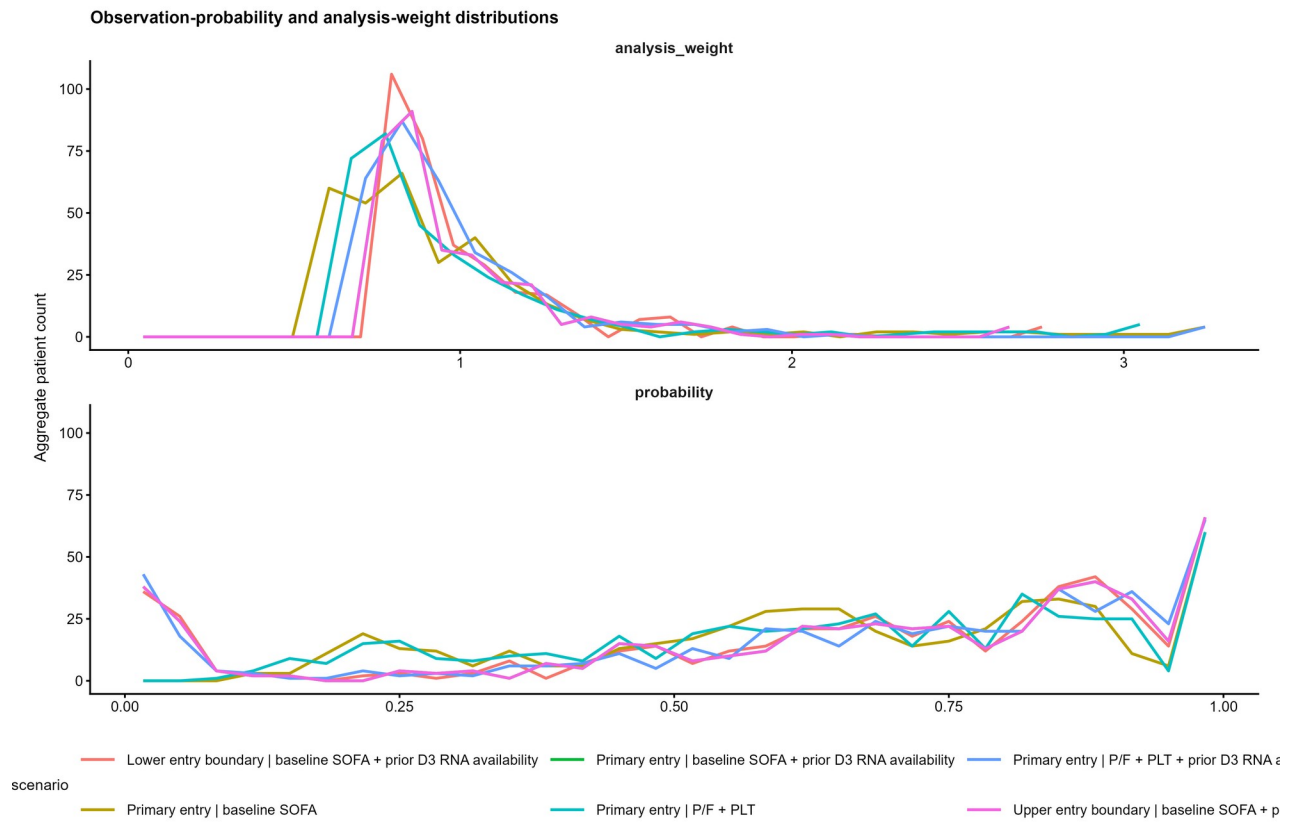
Supplementary Figures

Supplementary Figs. S1–S9 are presented in numerical order. Supplementary Figs. S1–S8 provide prespecified Day-5 RNA-seq availability and weighting diagnostics, Hallmark robustness analyses, entry-boundary sensitivity and the descriptive Day-5 plasma protein availability audit. Supplementary Fig. S9 presents the complete displayed clinical univariable Cox contrasts.

Centre-level Day-5 RNA-seq positivity support



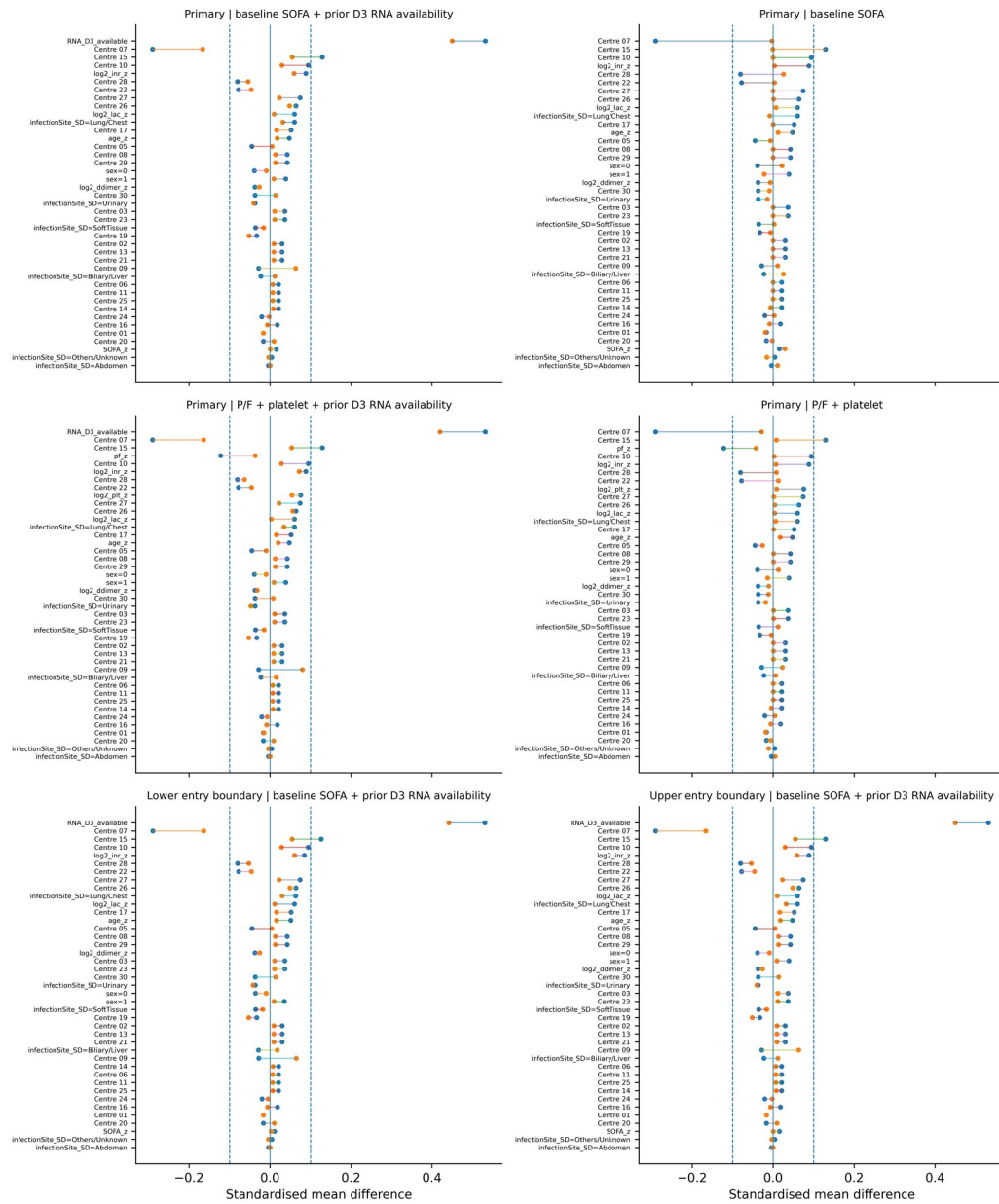
Supplementary Fig. S1 | Centre-level positivity support for Day-5 RNA-seq availability. Centre-level numbers of observed and unobserved Day-5 RNA-seq samples are shown for the prespecified primary entry definition. Centres were classified as zero-, partial- or all-observation centres before weighting. Patients from zero-observation centres were excluded from the positivity-supported estimand.



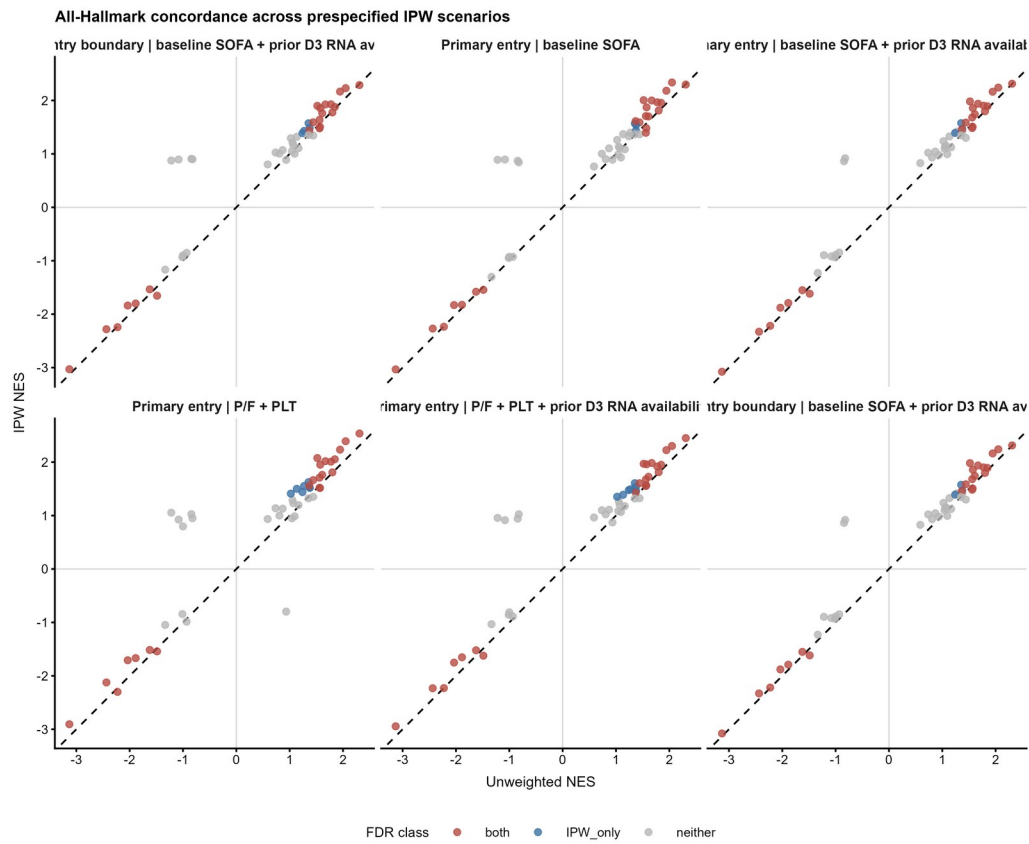
Supplementary Fig. S2 | Availability probabilities and inverse-probability weights. Aggregate binned distributions of model-estimated Day-5 RNA-seq observation probabilities and the corresponding non-stabilised inverse-observation-probability weights are shown for all six prespecified scenarios. Weights were truncated according to the prespecified scenario rules and normalised to mean one among observed patients.

Covariate balance before and after inverse-observation weighting

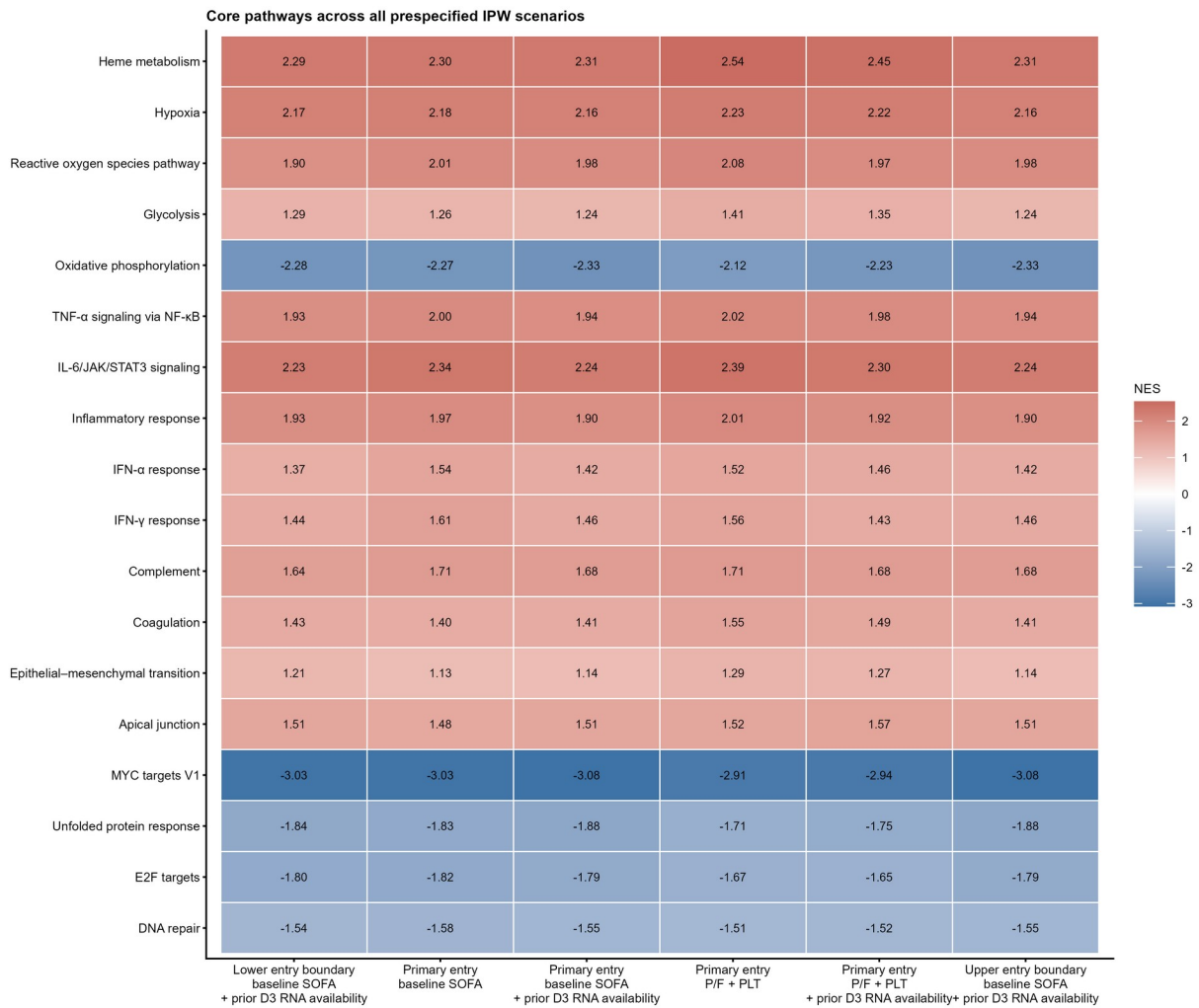
• Before weighting • After weighting



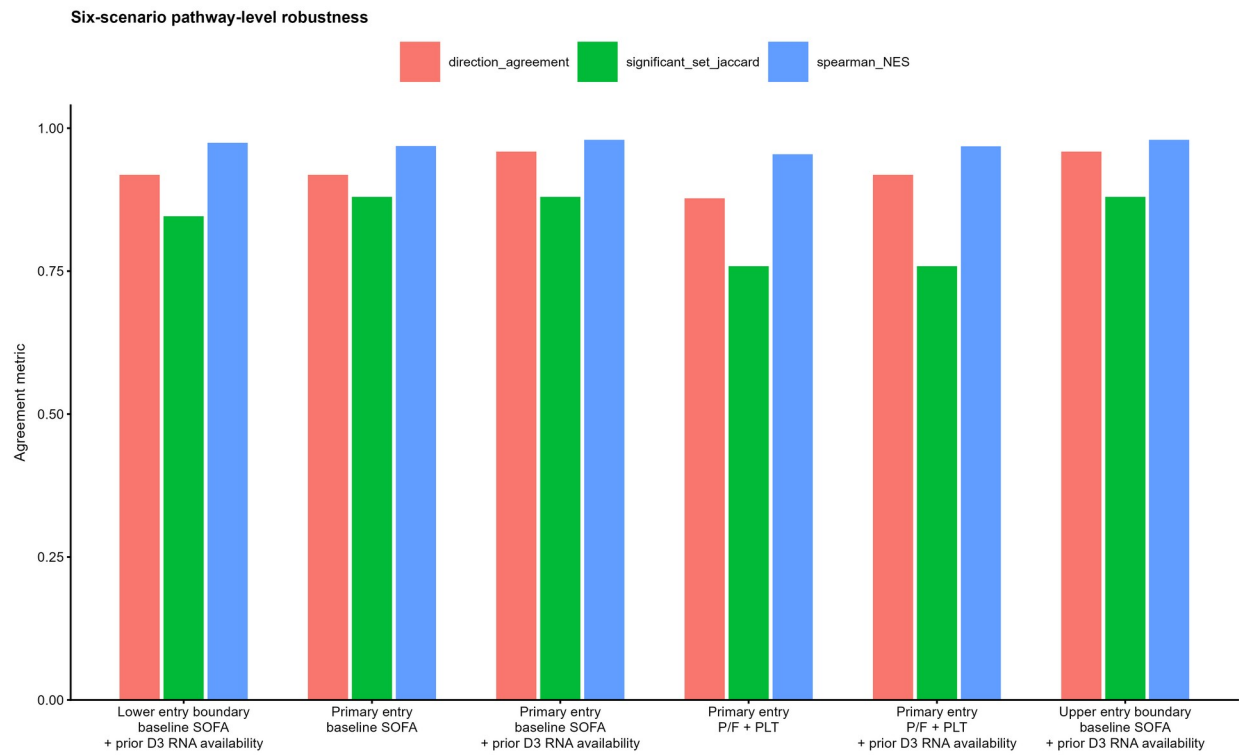
Supplementary Fig. S3 | Covariate balance before and after weighting. Signed standardised mean differences compare observed and unobserved Day-5 RNA-seq groups before weighting and after application of the prespecified primary inverse-probability weights. The vertical reference lines mark an absolute standardised mean difference of 0.10. Residual imbalance, particularly in prior Day-3 RNA-seq availability, remained after weighting.



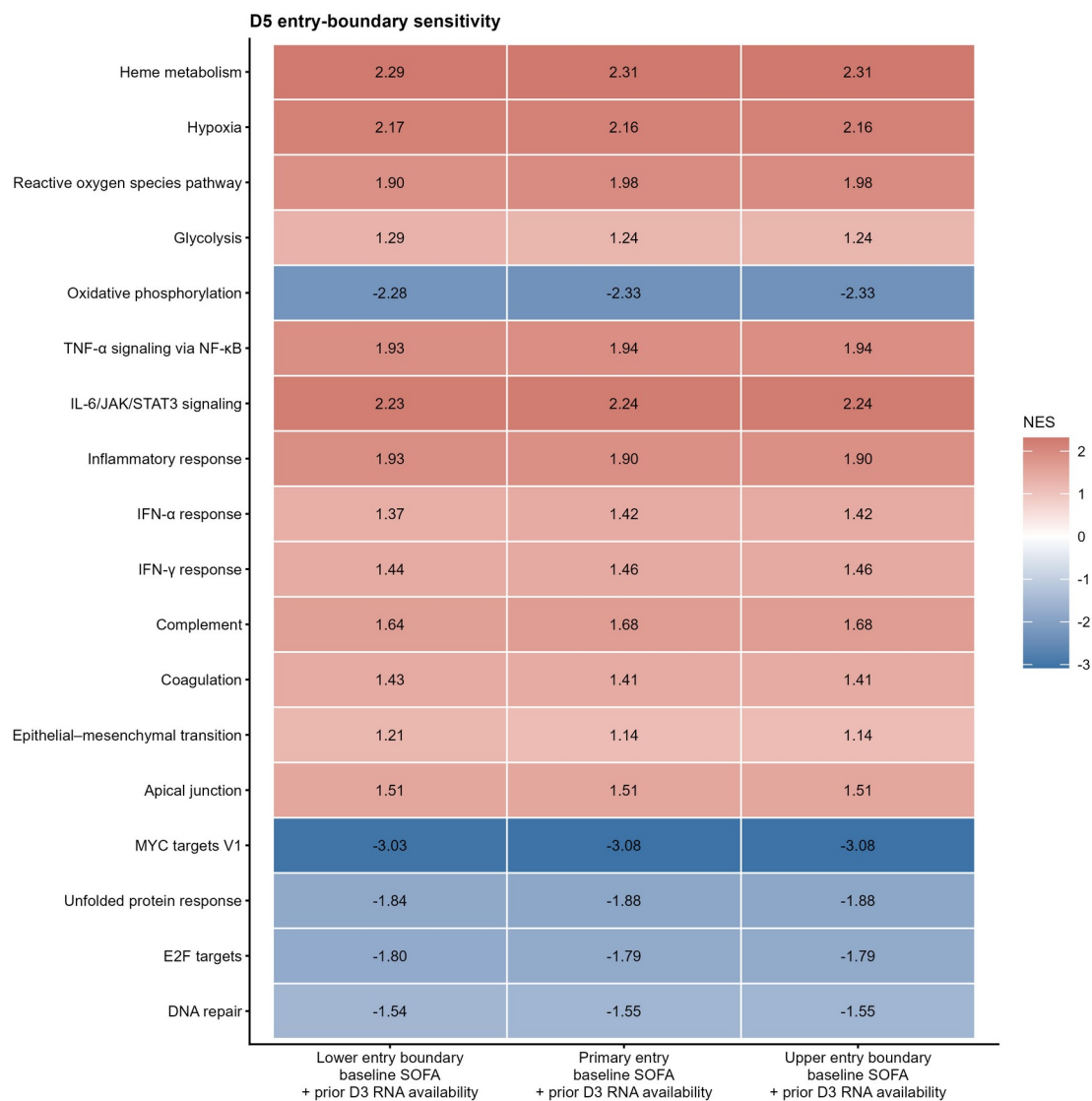
Supplementary Fig. S4 | Comparison of unweighted and IPW Hallmark enrichment. Each point compares Day-5 mortality-association NES from the unweighted centre-stratified primary analysis with NES from a prespecified weighted scenario across the complete Hallmark family. Colour denotes the joint FDR classification based on BH-FDR <0.05. The primary weighted scenario showed an NES Spearman correlation of 0.980, direction agreement of 95.9% and significant-set Jaccard index of 0.88 relative to the unweighted analysis. Concordance supports robustness of the overall pathway pattern but does not remove availability bias.



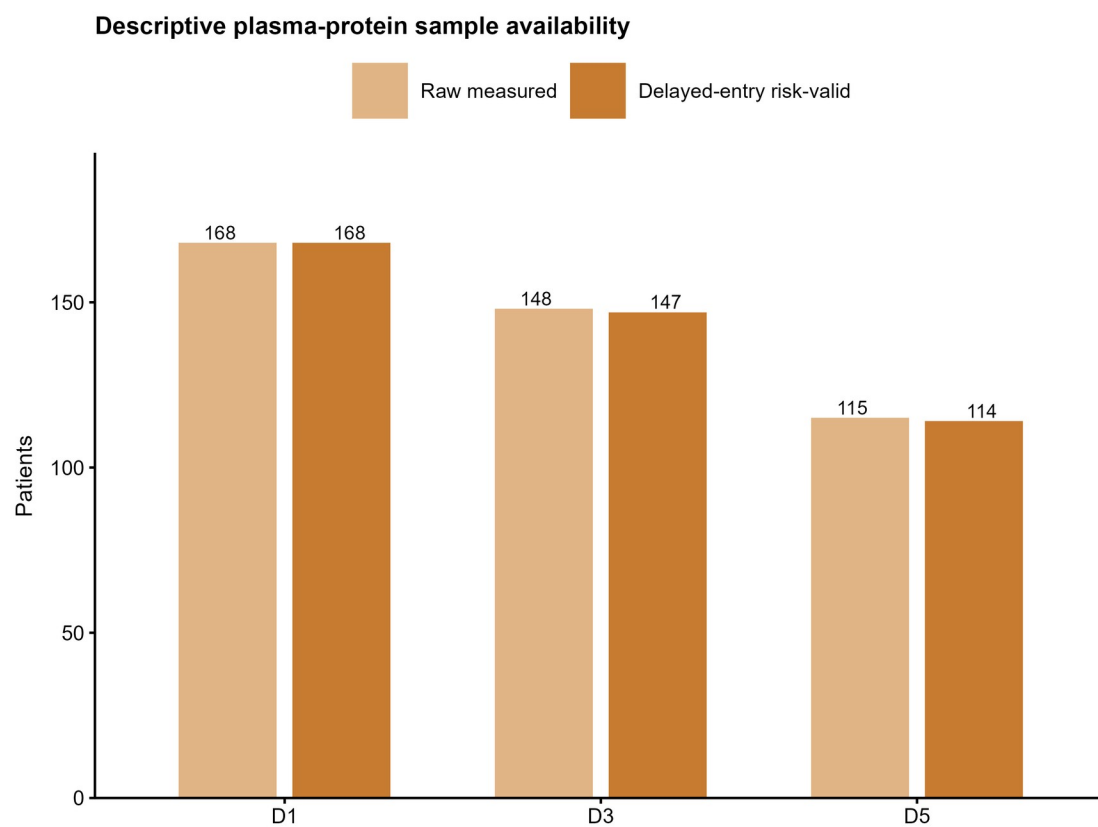
Supplementary Fig. S5 | Core pathways across prespecified IPW scenarios. Heat-map cells show weighted Day-5 mortality-association NES for the prespecified core Hallmark pathways across all six availability scenarios. Positive NES indicates enrichment among genes whose higher expression was associated with higher subsequent 60-day mortality risk. Negative NES indicates enrichment among genes whose higher expression was associated with lower subsequent mortality or, equivalently, whose lower expression was associated with higher mortality. NES does not denote absolute pathway expression. *BH-FDR<0.05, **BH-FDR<0.01 and ***BH-FDR<0.001 for the displayed scenario.



Supplementary Fig. S6 | Scenario-level Hallmark robustness metrics. Spearman correlation of NES, pathway-direction agreement and significant-set Jaccard overlap compare each weighted scenario with the unweighted Day-5 primary analysis across the complete Hallmark family.



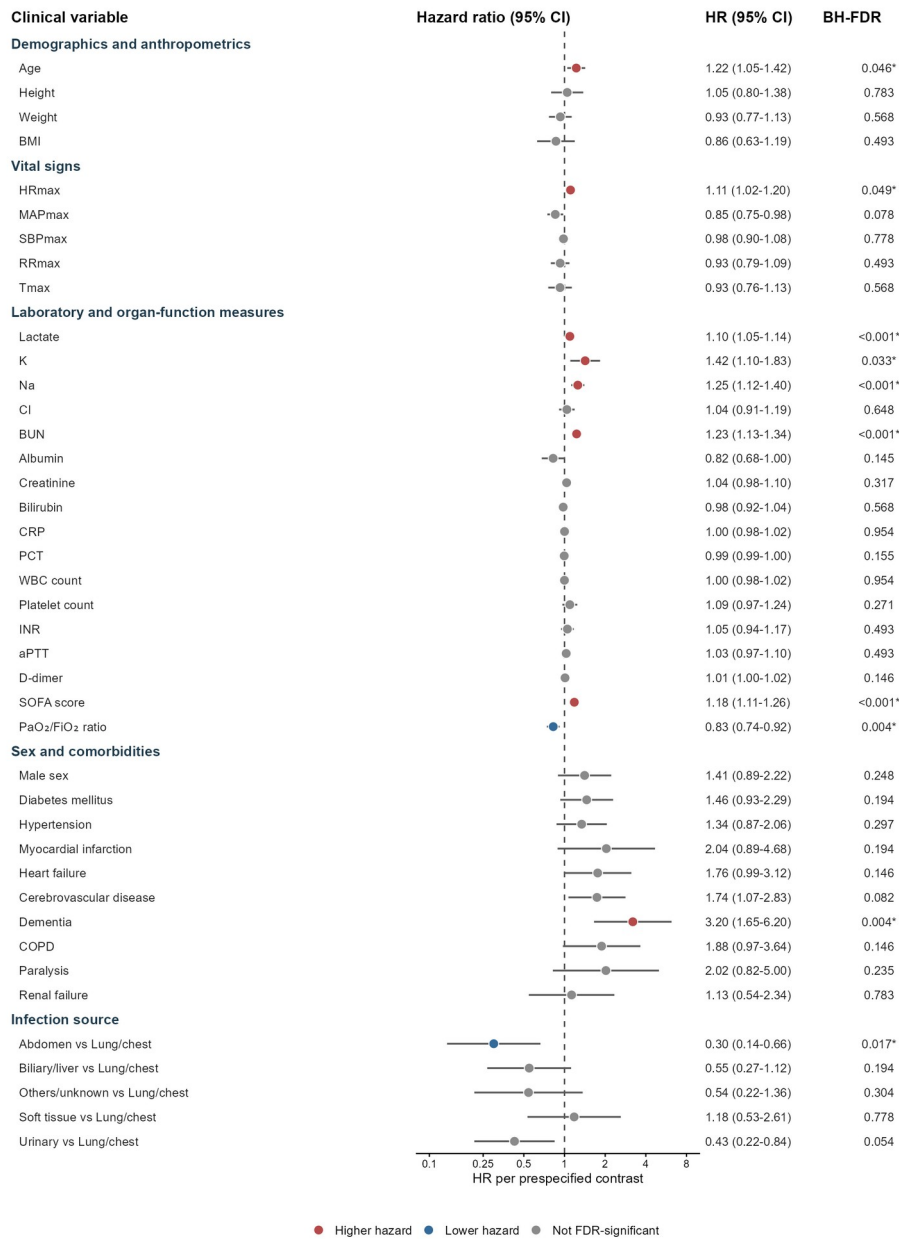
Supplementary Fig. S7 | Day-5 entry-boundary sensitivity. Core-pathway mortality-association NES values are compared across the prespecified lower, primary and upper Day-5 entry boundaries. Positive NES indicates enrichment among genes whose higher expression was associated with higher subsequent 60-day mortality risk. Negative NES indicates enrichment among genes whose higher expression was associated with lower subsequent mortality or, equivalently, whose lower expression was associated with higher mortality. *BH-FDR<0.05, **BH-FDR<0.01 and ***BH-FDR<0.001 for the displayed scenario.



Supplementary Fig. S8 | Descriptive availability of plasma proteomic samples across landmarks. Raw measured and delayed-entry risk-valid plasma protein sample counts are shown across Days 1, 3 and 5. This analysis is descriptive; no protein availability weights were estimated.

Supplementary Figure A9 | Exploratory clinical univariable Cox analysis

Higher or lower subsequent 60-day mortality hazard per prespecified contrast



Separate univariable Cox models; N=504 and 84 deaths. Variable labels follow Table 1; contrasts and units are reported in Table S2.

*BH-FDR<0.05; †nominal proportional-hazards diagnostic P<0.05; ‡nonlinearity likelihood-ratio P<0.05. Infection-source overall likelihood-ratio P=0.0038 (lung/chest reference). These estimates are descriptive, unadjusted and non-causal.

Supplementary Fig. S9 | Exploratory clinical univariable Cox associations with 60-day mortality. Points and horizontal bars show hazard ratios and 95% confidence intervals from separate univariable Cox models in 504 patients with 84 deaths. Variable names and abbreviations follow Table 1; prespecified contrasts and units are reported in Supplementary Table S2. The Benjamini–Hochberg procedure was applied across all 41 displayed contrasts; * indicates BH-FDR<0.05. † denotes nominal evidence against the proportional-hazards assumption and ‡ denotes nominal evidence of nonlinearity. Infection source was additionally associated with outcome in an overall likelihood-ratio test (P=0.0038). These univariable estimates are descriptive and are not mutually adjusted or causal.

Supplementary Tables

Supplementary Tables S1–S8 are supplied as separate machine-readable Excel workbooks. Supplementary Table S9 is included below and supplied as a separate Excel workbook.

Item	Title	Contents
Supplementary Table S1	Clinical variable definitions	Variable definitions, prespecified increments, transformations, infection-source harmonisation and missing-variable exclusions.
Supplementary Table S2	Clinical univariable Cox models	Complete exploratory univariable Cox contrasts, proportional-hazards diagnostics and nonlinearity tests.
Supplementary Table S3	RNA-seq gene-wise Cox and PH outputs	Complete gene-level Day-1, Day-3 and Day-5 centre-stratified Cox results and PH diagnostics.
Supplementary Table S4	RNA-seq Hallmark enrichment	Complete Hallmark enrichment and leading-edge results for transcriptomic analyses.
Supplementary Table S5	Protein-wise Cox and PH outputs	Complete protein-level Day-1, Day-3 and Day-5 centre-stratified Cox results and PH diagnostics.
Supplementary Table S6	Protein Hallmark enrichment	Complete Hallmark enrichment and leading-edge results for proteomic analyses.
Supplementary Table S7	Cross-omic models	Contemporaneous, forward, reverse-order and sensitivity cross-omic pathway models.
Supplementary Table S8	Day-5 availability and IPW	Positivity, observation-model, weight, balance and weighted-enrichment diagnostics.
Supplementary Table S9	Complete baseline characteristics	Complete Day-1 baseline characteristics according to 60-day outcome; this is the full source table from which main-text Table 1 was selected.

Supplementary Table S9. Complete baseline characteristics of the Day-1 SIC cohort according to 60-day outcome

Characteristic	Overall (n=504)	Survivor/censored (n=420)	Death (n=84)	SMD	P value
Age	68.00 [58.00, 78.00]	67.00 [56.00, 77.00]	72.00 [63.75, 81.00]	0.311	0.005
Height	166.00 [159.00, 170.00]	165.50 [159.00, 170.00]	167.50 [158.75, 171.00]	0.040	0.655
Weight	63.00 [55.00, 70.00]	63.00 [55.00, 70.00]	64.50 [55.00, 70.00]	-0.087	0.751
BMI	23.24 [20.81, 25.02]	23.28 [20.94, 25.08]	23.08 [20.76, 24.76]	-0.102	0.408
HRmax	119.00 [103.00, 135.00]	118.00 [102.00, 134.00]	121.00 [108.00, 139.25]	0.267	0.049
MAPmax	94.85 [84.00, 106.00]	95.00 [85.00, 107.00]	92.00 [82.00, 102.25]	-0.283	0.045
SBPmax	141.00 [126.00, 156.25]	141.00 [126.00, 156.00]	138.00 [125.25, 159.00]	-0.033	0.887
RRmax	26.00 [22.00, 32.00]	26.00 [23.00, 32.00]	26.00 [22.00, 32.00]	-0.103	0.347
Tmax	37.70 [37.00, 38.50]	37.70 [37.00, 38.50]	37.60 [36.77, 38.42]	-0.084	0.366
Lactate	2.80 [1.70, 5.20]	2.67 [1.61, 4.90]	3.88 [1.90, 6.90]	0.411	0.004
K	3.80 [3.35, 4.21]	3.80 [3.32, 4.20]	3.92 [3.43, 4.44]	0.303	0.044
Na	137.05 [133.50, 142.00]	137.00 [133.00, 141.10]	138.45 [135.00, 145.10]	0.364	0.032
Cl	107.00 [102.00, 112.00]	107.20 [102.00, 112.00]	106.05 [100.75, 112.70]	0.057	0.589
BUN	11.89 [8.10, 18.20]	11.26 [7.88, 16.52]	18.20 [9.80, 25.36]	0.530	<0.001
Albumin	28.25 [24.80, 31.63]	28.40 [25.00, 31.90]	27.25 [24.15, 29.75]	-0.238	0.038
Creatinine	134.55 [93.00, 223.37]	130.30 [91.75, 217.50]	163.75 [112.82, 247.43]	0.154	0.006
Bilirubin	23.00 [13.30, 42.70]	23.00 [13.67, 43.71]	21.40 [12.17, 37.08]	-0.099	0.422
CRP	153.95 [79.64, 219.64]	154.61 [77.35, 226.05]	147.84 [83.68, 206.16]	0.001	0.897
PCT	23.05 [5.22, 86.87]	24.40 [5.74, 91.32]	12.25 [2.20, 57.23]	-0.221	0.026
WBC	11.60 [7.11, 17.68]	11.59 [7.19, 17.18]	12.00 [5.83, 19.92]	-0.003	0.892
Platelet	89.00 [57.00, 132.25]	86.50 [55.00, 131.00]	96.50 [68.00, 143.25]	0.168	0.064
INR	1.38 [1.23, 1.61]	1.37 [1.22, 1.57]	1.49 [1.31, 1.65]	0.119	0.007
aPTT	37.60 [32.10, 45.00]	37.25 [31.75, 44.75]	39.35 [34.08, 45.85]	0.107	0.133
D-dimer	6.06 [2.73, 14.95]	5.98 [2.73, 14.59]	7.07 [2.75, 18.14]	0.187	0.584
SOFA score	10.00 [7.00, 12.00]	9.00 [7.00, 12.00]	12.00 [10.00, 14.00]	0.636	<0.001
PF	243.82 [168.06, 327.52]	249.70 [178.38, 330.45]	205.30 [122.64, 280.95]	-0.404	0.001
Sex, male, n (%)	298 (59.1%)	242 (57.6%)	56 (66.7%)	0.187	0.145
Diabetes mellitus	137 (27.2%)	108 (25.7%)	29 (34.5%)	0.193	0.107
Hypertension	200 (39.7%)	161 (38.3%)	39 (46.4%)	0.164	0.180
Myocardial infarction	20 (4.0%)	14 (3.3%)	6 (7.1%)	0.172	0.122
Heart failure	53 (10.5%)	39 (9.3%)	14 (16.7%)	0.221	0.052
Cerebrovascular disease	88 (17.5%)	66 (15.7%)	22 (26.2%)	0.260	0.027
Dementia	23 (4.6%)	13 (3.1%)	10 (11.9%)	0.339	0.002
COPD	35 (6.9%)	25 (6.0%)	10 (11.9%)	0.210	0.060
Paralysis	16 (3.2%)	11 (2.6%)	5 (6.0%)	0.165	0.161
Renal failure	42 (8.3%)	34 (8.1%)	8 (9.5%)	0.050	0.666
Infection source, n (%)					
Abdomen	90 (17.9%)	83 (19.8%)	7 (8.3%)	-0.333	0.003
Biliary/liver	66 (13.1%)	57 (13.6%)	9 (10.7%)	-0.088	
Lung/chest	194 (38.5%)	148 (35.2%)	46 (54.8%)	0.400	
Others/unknown	36 (7.1%)	31 (7.4%)	5 (6.0%)	-0.057	
Soft tissue	25 (5.0%)	18 (4.3%)	7 (8.3%)	0.167	
Urinary	93 (18.5%)	83 (19.8%)	10 (11.9%)	-0.216	

Data are median [interquartile range] or n (%). P values are two-sided comparisons between survivors/censored patients and patients who died within 60 days. Continuous variables were compared using Wilcoxon rank-sum tests and categorical variables using Fisher's exact tests. For infection source, the P value shown in the first category row is the global P value for the six-level variable. P values were descriptive and were not used for covariate selection. aPTT, activated partial thromboplastin time; BMI, body mass index; BUN, blood urea nitrogen; Cl, chloride; COPD, chronic obstructive pulmonary disease; CRP, C-reactive protein; HRmax, maximum heart rate; INR, international normalised ratio; K, potassium; MAPmax, maximum mean arterial pressure; Na, sodium; PCT, procalcitonin; PF, PaO₂/FiO₂ ratio; RRmax, maximum respiratory rate; SBPmax, maximum systolic arterial pressure; SIC, sepsis-induced coagulopathy; SMD, standardised mean difference; SOFA, Sequential Organ Failure Assessment; Tmax, maximum temperature; WBC, white blood cell.